Final Project Report
Data Warehouse

Bachelor of Science in Data Science, Faculty of Mathematics, Science, and Natural Sciences
Surabaya State University

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URL Github	https://github.com/JullMol/NewsFlow

General Project Information

Title	Implementation of an ETL Pipeline-Based Data Warehouse at Kompas.com News with Sentiment Analysis, Trending Topics, and Multidimensional OLAP Visualization
Topic	Social / Media & Communication
Revision?	Yes / No

Description of the Research Question	Kompas.com is one of the largest online news portals in Indonesia, producing hundreds of articles per day covering national, economic, technology, entertainment, and sports categories. Although the volume of this data is enormous and continues to grow daily, there is currently no structured system in place to aggregate, clean, and store this news data in a format ready for longitudinal or multidimensional analysis. This issue is urgent because the potential information contained in the Kompas.com news archive-ranging from patterns of public sentiment, the emergence of figures and issues, to topic trends-has not yet been systematically utilized for analytical purposes. Some of the main challenges causing this issue to remain unresolved include: **1). Unstructured and scattered data**: News content consists of free-form text scattered across thousands of web pages, requiring a reliable web scraping-based data acquisition mechanism using XML sitemaps. **2). No official public API available**: Kompas.com does not provide API endpoints for programmatic data access, so the entire acquisition process must be designed independently while taking into account the stability of the HTML structure. **3). The complexity of NLP pipelines for Indonesian**: Sentiment analysis, Named Entity Recognition (NER), and vector embedding-based semantic representation for Indonesian-language content require tailored models (such as IndoBERT). **4). Integration into a Data Warehouse architecture**: Incorporating NLP processing results into a multidimensional schema (star schema) that supports OLAP mechanisms requires a comprehensive ETL pipeline design, from extraction to loading into an Atoti-based DataMart.

The target insights to be achieved	Through the CUBE query mechanism in Atoti DataMart, the insights we aim to gain include: ● New volume trends: Changes in the number of articles per category per month over a 1-2 year period, to identify which categories are experiencing consistent growth or decline in content production. ● Sentiment trends over time:Sentiment distribution (positive/neutral/negative) by category and time period, enabling the identification of shifts in the tone of coverage regarding specific issues. ● Analysis of dominant entities: The most frequently mentioned figures, institutions, and locations by category and by period, utilizing NER results to identify who or what dominates the news coverage the most. ● Daily trending topic detection: Identification of topics experiencing a significant spike in keyword frequency compared to their historical average, as an indicator of emerging issues. ● Cross-temporal semantic similarity: Use of vector embedding to measure semantic similarity between articles and identify recurring content patterns across categories and time periods.
Research References	https://doi.org/10.1371/journal.pone.0276367 https://doi.org/10.1016/j.procs.2023.12.051 https://doi.org/10.7717/peerj-cs.1715 https://doi.org/10.1016/j.asoc.2023.111112

Dataset Details

Data Source	The data was obtained through web scraping from the Kompas.com news portal (https://www.kompas.com) using a crawling approach based on monthly XML sitemaps. This process aligns with the approach used by Pichiyan et al. (2023), who demonstrated that NLP-based web scraping is a reliable technique for acquiring unstructured text from web sources when public APIs are unavailable. Historical sitemap archives were collected for a 1-2-year timeframe to build the base dataset, followed by an automated daily acquisition pipeline that runs periodically using an Airflow DAG to keep the dataset up to date.
General Description of the Dataset/Data Dump	The dataset consists of a collection of daily news articles from Kompas.com gathered through web scraping over the past 1-2 years, with an estimated total of thousands of articles. The dataset covers various news categories, such as national, economy, technology, health, sports, and politics, allowing it to provide a longitudinal representation of the dynamics of information and public issues in Indonesia. Each article is stored in a structured format with key attributes including the URL, article title, full news content, author name, category, sub-category, topic tags, publication date and time, last updated date and time, scraping timestamp, article word count, extracted content status, and article validity status. The dataset is also enriched using a Natural Language Processing (NLP) approach based on the Transformer and IndoBERT models. This process generates additional attributes, including sentiment analysis indicating the article's sentiment polarity (positive, neutral, or negative), as well as an 'embedding_vector' representing the article's semantic meaning in the form of a high-dimensional numerical vector. The embedding representation enables analysis of contextual similarity between articles and supports various text-based machine learning applications. The dataset also includes NER-derived entities to detect important

	entities within articles, such as names of figures, organizations, institutions, companies, and locations.
Specific Aspects of The Dataset	1. Sparsity The data is sparse in the tags and author columns, as some articles lack explicit tags or do not list an author. This sparsity is common in news data collected via web scraping, as text from web sources often lacks consistent metadata completeness. Estimates of the sparsity level in both columns will be calculated during the preprocessing stage. 2. Imbalanced Data The dataset is imbalanced across categories. Categories such as national news and economics tend to have a much larger volume of articles compared to categories like technology, entertainment, or sports. A similar situation was found in the study by Murfi et al. (2024) on an Indonesian text dataset, where class imbalance was a major challenge that had to be addressed through techniques such as oversampling or class weighting prior to training the sentiment model. The imbalance ratio between categories will be measured empirically during the initial data exploration (EDA) phase. 3. Outliers & Article Length Distribution The distribution of article length (measured by the number of words in the content column) varies significantly across the dataset. Some articles are very short, such as breaking news updates or photo captions, while others are much longer and contain detailed in-depth reporting. These differences create outliers in the dataset because the text length between articles can differ substantially. Variations in article length may affect the performance of Natural Language Processing (NLP) models, especially during text representation and feature extraction. Short articles may contain limited contextual information, while excessively long articles may introduce noise and increase processing complexity. Similar challenges related to text-length variation in news datasets have also been discussed by Rozado et al. (2022), where normalization and consistent preprocessing were important before applying Transformer-based models. To handle this issue, text normalization and preprocessing will be applied during the ETL (Extract, Transform, Load) stage.

Data Staging Results

Extraction Phase	The data extraction process is carried out in two stages. First, article URLs are collected in bulk by parsing Kompas.com's XML sitemap index, which contains sub-sitemaps with hundreds of article URLs per month. The system limits the collection to a maximum of 50 articles per day, selected at random, to ensure a balanced sample; if a date is not covered in the sitemap, the system automatically falls back to the Kompas index page (indeks.kompas.com). Once the URLs have been collected, the articles are scraped to extract key components, including the URL, title, main content, author name, category, tags/keywords, publication date, scraping time, and word count. The category is derived from the URL structure, while the publication date is retrieved from the meta tag, with a fallback to the URL pattern if the meta tag is unavailable. Each successfully scraped article is saved to a daily JSON file as a temporary file before proceeding to the transformation stage.

The raw data obtained from scraping is stored in a daily JSON format before entering the transformation phase. Each article is represented as a single JSON object with raw attributes extracted directly from the Kompas.com webpage. Here is an example of a raw article obtained from scraping:

Table 1. Attributes of Raw Data from Scraping

Attribute	Example Value	Description
url	https://internasional.kompas.com/read/2026/05/13/224353370/trump-boyong-18-ceo-ke-china-mau-bertemu-xi-jinping	Article source URL extracted from the XML sitemap.
title	Trump Boyong 18 CEO ke China, Mau Bertemu Xi Jinping	Raw article title extracted from HTML tags.
content	BEIJING, KOMPAS.com - Presiden Amerika Serikat Donald Trump membawa 18 CEO...	Raw article content, still containing boilerplate text such as "KOMPAS.com –", "Baca juga:", and similar elements.
author	(BRENDAN SMIALOWSKI)	Author or photographer name, not yet normalized.
category	Read	Raw category derived from the URL structure, not yet reclassified.
tags	["China", "Donald Trump", "trump", "trump bawa 18 ceo ke china", ...]	Tags/keywords may contain duplicates and have not yet been deduplicated.
pub_date	2026-05-13	Article publication date obtained from HTML meta tags.
scraped_at	2026-05-14T12:19:32.376561+00:00	Timestamp indicating when the article was scraped.
word_count	242	Word count calculated from the raw content.

At this raw stage, NLP-derived attributes such as sentiment_label, sentiment_score, embedding_vector, and entities (PERSON, ORGANIZATION, LOCATION) do not yet exist. These attributes are only added after the article passes through the Transformation Phase. Additionally, the tags field may contain duplicate values and the category field is not yet normalized, as both are extracted directly from the raw HTML structure without further cleaning.

Transformation Phase

The transformation is carried out in 6 sequential steps, as follows:

1. Text cleaning, where the article content is stripped of HTML tags, special encoding characters, and boilerplate text typical of Kompas.com, such as the phrases “Read also:”, “KOMPAS.com –”, and “Get news updates...”. Articles with less than 20 words are discarded, and duplicate URLs are eliminated at this stage.
2. Sentiment analysis using the IndoBERT model. The model’s input consists of the title combined with the first 500 characters of the article’s content, producing a sentiment label (positive, neutral, or negative) along with a confidence score.
3. Embedding generation, which produces a 384-dimensional vector per article using a Sentence Transformer model based on MiniLM-L12 (paraphrase-multilingual-MiniLM-L12-v2) to represent the article’s semantic meaning.
4. Performing Named Entity Recognition (NER) to identify three types of entities from the article content: people (PERSON), institutions/organizations (ORGANIZATION), and locations/regions (LOCATION).
5. The NER results are then enriched through a Named Entity Linking (NEL) process, which links each entity to external knowledge bases such as Wikidata and Wikipedia to obtain additional attributes like entity descriptions and similarity scores from the matching results.
6. Detects daily trending topics by calculating keyword occurrence frequency and comparing it to historical averages to generate a trending score per keyword per date.

After completing all six transformation steps, the processed data is ready to be loaded into the data warehouse. The following table presents an example of a single article record after the full ETL transformation, using the same article from the raw data example above for direct comparison:

Table 2. Example of Processed Article Data After ETL Transformation

Attribute	Example Value	Description
artikel_id	15399	Primary key, automatically assigned when the record is loaded into PostgreSQL.
url	https://internasional.kompas.com/read/2026/05/13/224353370/trump-boyong-18-ceo-ke-china-mau-bertemu-xi-jinping	Validated and deduplicated article URL.
judul	Trump Boyong 18 CEO ke China...	Article title retained after validation.
konten	BEIJING, Presiden Amerika Serikat...	Cleaned article content with boilerplate text (e.g., "KOMPAS.com –", "Baca juga:") removed.
waktu_id	130	Foreign key referencing the **dim_waktu**

			(publication date) dimension.
	kategori_id	511	Foreign key referencing the **dim_kategori** dimension.
	penulis_id	2	Foreign key referencing the **dim_penulis** dimension.
	sentimen_id	2	Foreign key referencing the **dim_sentimen** dimension (2 = neutral).
	sentimen_score	0.0	Polarity score represents both direction and confidence. Ranges from -1.0 (strong negative) to +1.0 (strong positive), with neutral sentiment mapped to 0.0 for mathematically valid averaging.
	embedding	[0.0790, 0.0254, 0.0425, ..., -0.0246]	384-dimensional semantic vector generated using the Sentence Transformer IndoBERT model.
	jumlah_kata	242	Word count calculated after content cleaning.
	tanggal_publikasi	2026-05-13	Normalized article publication date.
	created_at	2026-05-14 12:40:21	Timestamp indicating when the record was loaded into the data warehouse.
	persons	["Amerika Serikat Donald Trump", "Trump"]	Person entities extracted using Named Entity Recognition (NER) and linked to Wikidata through Named Entity Linking (NEL) (e.g., Q22686 = Donald Trump).
	organizations	[]	Organization entities extracted using NER. No organization entities were detected in this article.

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<tr><td>locations</td><td>["AS", "China", "Amerika Serikat"]</td><td>Location entities extracted using NER and enriched with additional information from Wikipedia/Wikidata through NEL.</td></tr>
</table>
The following is a NER/NEL sub-table showing the enrichment details:
Table 2a. Example of NEL Enrichment Result for Extracted Entities
<table>
<tr><th>Entity Name</th><th>Type</th><th>Count</th><th>Wikidata ID</th><th>NEL Matched</th><th>NEL Score</th><th>NEL Similarity</th></tr>
<tr><td>AS</td><td>LOCATION</td><td>5</td><td>Q190580</td><td>True</td><td>0.75</td><td>1.00</td></tr>
<tr><td>China</td><td>LOCATION</td><td>4</td><td>Q148</td><td>True</td><td>0.75</td><td>1.00</td></tr>
<tr><td>Amerika Serikat</td><td>LOCATION</td><td>1</td><td>Q30</td><td>True</td><td>1.00</td><td>1.00</td></tr>
<tr><td>Trump</td><td>PERSON</td><td>1</td><td>Q22686</td><td>True</td><td>1.00</td><td>0.95</td></tr>
</table>
Compared to the raw data in Table 1, the processed record demonstrates several key transformations: boilerplate content has been cleaned, raw strings have been replaced with normalized foreign keys, NLP-derived attributes (sentimen_score, embedding) have been generated, and named entities (persons, organizations, locations) have been extracted and linked to external knowledge bases (Wikidata/Wikipedia) through the NEL process.
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<td>Load Phase</td>
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The transformed data is loaded into PostgreSQL using the UPSERT pattern to ensure idempotency, so that the pipeline can be rerun without creating duplicate data. The loading process includes populating dimension tables, such as the time dimension, which is automatically populated from the publication date with day, week, month, quarter, and year attributes; the category dimension; the author dimension; the sentiment dimension, which is pre-populated with three labels (positive, neutral, negative); and the entity dimension, supported by NEL result attributes. The main fact table is partitioned by publication date per month from 2024 to 2026, equipped with indexes on key columns and a trigram index on the title column for fast text searches. Additionally, the bridge table stores the relationship between articles and entities along with their frequency of occurrence, while the trending fact table stores trending keywords along with their scores per date. After each batch is fully loaded, three materialized views are automatically refreshed to support efficient OLAP queries: article aggregation by category per month, daily sentiment trends, and a list of the most dominant entities. A savepoint mechanism is also implemented for error isolation per article, ensuring that a failure in a single article does not invalidate the entire batch.
Based on monitoring results for PostgreSQL on Supabase, the current database size is approximately 94 MB. This size includes scraped article data, dimension tables, fact tables, indexes, table partitions, and materialized views used in the analytics process. As data ingestion and NLP processing activities increase, the database size is expected to continue growing gradually.
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To verify the loaded data, the following shows a sample of 10 article records retrieved directly from the fact_artikel table in Supabase, joined with the dim_kategori and dim_sentimen dimension tables:

artikel_id	judul	nama_kategori	label	sentimen_scor	jumlah_kata	tanggal_publi
15895	Klasemen MotoGP 2026 Usai Sprint Race Italia	Read	positive	0.5646	140	2026-05-30
15894	KA Cikuray Terus Tumbuh, Hubungkan Garut dan Jakarta dengan Tarif Te	Read	positive	0.9839	388	2026-05-30
15893	Timwas Haji DPR Minta Masalah Fasilitas di Mina Segera Diatasi	Read	neutral	0	521	2026-05-30
15892	Wamenhaj: Fase Armuzna Tuntas, Seluruh Jemaah Haji Indonesia Telah T	Read	neutral	0	260	2026-05-30
15891	Israel Kuasai 60 Persen Gaza, Netanyahu Targetkan 70 Persen	Tren	neutral	0	247	2026-05-30
15890	Sering Tak Disadari Penyelam, Kebiasaan Ini Bisa Rusak Terumbu Karan	Read	positive	0.5014	421	2026-05-30
15889	Bengkel di Cikupa Tangerang Kebakaran, 1 Orang Tewas Terjebak di Toi	Read	neutral	0	135	2026-05-30
15888	Harga Jay Idzes Melonjak usai Jadi Pilihan Utama Sassuolo di Serie A	Read	neutral	0	210	2026-05-30

Table 3. Sample Query Result from fact_artikel (Supabase)

Column	Description
artikel_id	Unique identifier for each article, automatically assigned by PostgreSQL.
judul	Article title after the content cleaning process.
nama_kategori	Resolved category name obtained from the category dimension table (dim_kategori).
label	Sentiment classification result (*positive*, *neutral*, or *negative*) obtained from the sentiment dimension table (dim_sentimen).
sentimen_score	Polarity score ranging from -1.0 (strong negative) to +1.0 (strong positive), where neutral sentiment is mapped to 0.0. This score reflects both the sentiment class and model confidence in a single continuous scale.
jumlah_kata	Total number of words in the article after the ETL cleaning process.
tanggal_publikasi	Normalized publication date of the article.

The PostgreSQL technology used includes table partitioning based on publication date ranges, B-Tree indexes on key columns, trigram indexes for text searches, the pgvector extension for vector embedding-based storage and search, and materialized views to accelerate OLAP analytical queries.

Performance benchmarking was conducted using the EXPLAIN ANALYZE feature in PostgreSQL. There were 4 benchmark scenarios:

Table 4. PostgreSQL Performance Benchmarking Results

Benchmark	Description	Without Optimization	With Optimization	Result

Materialized View	Comparing the execution times of queries for article aggregates by category and month directly on the fact and dimension tables versus queries using the materialized view 'mv_artikel_per_kategori_bulan'.	2230.928 ms	10.577 ms	Query execution using the materialized view achieved approximately **210.9x** faster performance compared to direct aggregation queries.
Partition Pruning	Measuring the efficiency of filter queries based on publication date ranges in the 'fact_artikel' table, which is partitioned by month.	-	1.059 ms	PostgreSQL successfully performed partition pruning with status Pruning Active: True, ensuring that only relevant partitions were scanned.
B-Tree Index	Comparing the performance of filter queries based on category and sentiment with and without the use of a B-Tree index on the 'category_id' and 'sentiment_id' columns.	10.565 ms	25.767 ms	In this scenario, the indexed query performed approximately **0.4x** slower, likely due to the relatively small dataset size where sequential scans remain more efficient.
pgvector Similarity	Measuring the time required to calculate cosine similarity (`<=>`) between article embeddings, with the option to create an IVFFlat index if the dataset contains ≥ 100 rows.	3687.209 ms	176.411 ms (IVFFlat)	The use of the IVFFlat index improved semantic similarity search performance by approximately **20.9x** faster compared to vector search without indexing.

Data Presentation Results

Subject Focus	The topic covered in this Multidimensional OLAP process is *Real-Time Analysis of Productivity, Topic Trends, and Public Opinion Sentiment in Kompas News*

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Articles. This OLAP process was formulated to address the challenges of the modern media industry in understanding information dissemination patterns and the dynamics of public opinion in a measurable manner. This analytical topic focuses on four main pillars:

1. Analyzes the level of publication activity of Kompas news articles across various categories/curricula to map editorial focus.
2. Tracks the emotional content of news articles (Labels: Positive, Negative, Neutral) across dimensions (time, category, and author) to measure journalistic objectivity.
3. Identifies daily topic explosions based on the frequency of keyword appearances as an indicator of current issues in society.
4. Analyzes the relationship between important figures, institutions, and geographic locations that dominate news coverage and the bias of surrounding opinion.

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<td>Star Schema</td>
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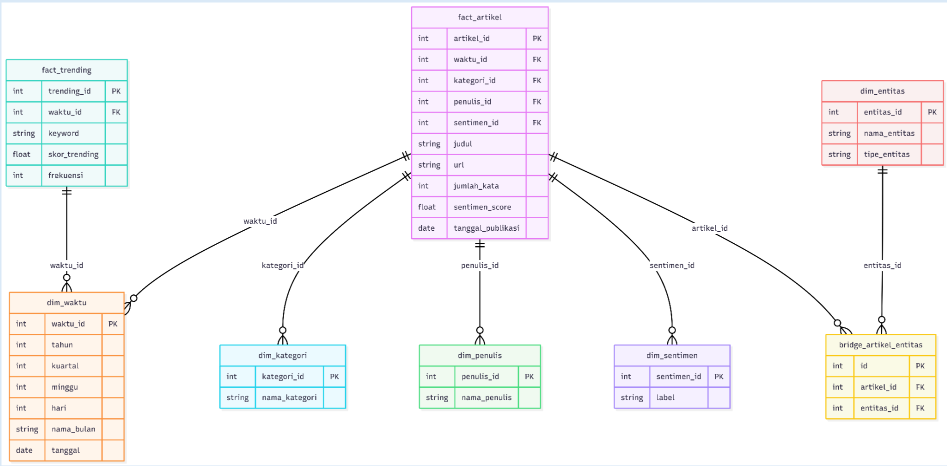

In the implementation of an OLAP architecture using the Atoti framework, the roles of each table within the Star Schema are structured into fact tables and dimension tables. The fact_artikel table serves as the primary fact table, storing the primary quantitative metrics for analysis with sentiment_score, which represents the emotional/opinion score of the news item, and word_count, which represents the depth of the article. As a supporting factor, the fact_trending table also serves as a secondary fact table, containing trending_score (strength of topic trends) and frequency (keyword repetition rate). Both fact tables contain a set of foreign keys that directly link them to the surrounding dimension tables, facilitating dynamic data aggregation calculations.

Furthermore, dimension tables provide attribute context (Hierarchies & Levels) that allows users to filter, group (Group by), and interactively navigate the data. The dim_time table plays a crucial role in forming a hierarchical Time Hierarchy (Year > Quarter > Month > Date), which underlies the interactive analytical techniques Drill-Down (seeing trend details from year to publication date details) and Roll-Up (grouping data from daily to monthly). The category dimension contained by the dim_category table provides a Category Hierarchy (Category Name), which is specifically involved in mapping and comparing the characteristics of news articles between sections (topical and regional).

To group articles based on emotional clusters, the dim_sentiment table provides a Sentiment Hierarchy (Sentiment labels: positive, neutral, negative) that divides the

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distribution of opinions in a measurable manner. Meanwhile, the dim_author table is involved in providing an Author Hierarchy (Author Name) to evaluate the productivity and writing style of each journalist.

Finally, the dim_entitas table, along with the bridge_artikel_entitas intermediate table, serves to bridge the Many-to-Many relationships between news stories and key entities (people, organizations, locations), enabling in-depth Named Entity Recognition (NER) correlation analysis of news opinion bias. This integrated combination of all fact tables and dimension tables creates a highly responsive and information-rich OLAP Cube foundation.

OLAP Result

To demonstrate the OLAP operations implemented in the Atoti cube, the following presents the terminal output of four OLAP operations executed against the 15,228 article records stored in the data warehouse.

OLAP Operation 1: Roll-Up (Tanggal → Bulan → Tahun per Kategori)

Roll-Up is an aggregation operation that summarizes data from a finer granularity to a coarser level along a dimensional hierarchy. In this operation, article volume data is progressively aggregated from the daily level (Tanggal) up to the monthly level (Bulan) and then to the yearly level (Tahun), grouped by news category. For example, the Badminton category recorded 3 articles in June 2024 at the monthly level, which when rolled up to the yearly level shows a total of 27 articles for the entire year of 2024. This operation is useful for identifying long-term publication trends across categories without being distracted by daily fluctuations.

[Granular] Article Volume per Date & Category (First 10 Rows):

Nama Kategori	Tahun	Kuartal	Bulan	Tanggal	Jumlah Artikel
Agri	2025	4	December	2025-12-09	1
Agri	2026	1	March	2026-03-12	1
Badminton	2024	2	June	2024-06-03	1
Badminton	2024	2	June	2024-06-15	2
Badminton	2024	2	May	2024-05-16	1
Badminton	2024	2	May	2024-05-19	2
Badminton	2024	3	August	2024-08-06	1
Badminton	2024	3	July	2024-07-12	2
Badminton	2024	3	July	2024-07-28	7
Badminton	2024	3	September	2024-09-01	2

[Roll-up to Bulan] Article Volume per Month & Category (First 10 Rows):

Nama Kategori	Tahun	Kuartal	Bulan	Jumlah Artikel
Agri	2025	4	December	1

Agri	2026	1	March	1
Badminton	2024	2	June	3
Badminton	2024	2	May	3
Badminton	2024	3	August	1
Badminton	2024	3	July	9
Badminton	2024	3	September	5
Badminton	2024	4	November	3
Badminton	2024	4	October	3
Badminton	2025	1	February	1

[Roll-up to Tahun] Article Volume per Year & Category:

Nama Kategori	Tahun	Jumlah Artikel
Agri	2025	1
Agri	2026	1
Badminton	2024	27
Badminton	2025	25
Badminton	2026	17
Bandung	2024	92
Bandung	2025	176
Bandung	2026	67
Banten	2025	5
Banten	2026	4

OLAP Operation 2: Drill-Down (Tahun 2026 → Bulan Mei)

Drill-Down is the reverse of Roll-Up, it navigates from a coarser level of granularity down to a finer level to reveal more detailed information. In this operation, the analysis starts from the yearly level (2026) and drills down to the monthly level to reveal article distribution per category throughout 2026. For example, the Bandung category shows 18 articles in January 2026, 16 in February, and 11 in March. This operation is useful for investigating which specific periods experienced spikes or drops in publication volume within a given year.

[Drill-Down to Bulan] Article Distribution per Category in Year 2026 (First 10 Rows):

Nama	Tahun	Kuartal	Bulan	Jumlah

Kategori				Artikel
Agri	2026	1	March	1
Badminton	2026	1	February	1
Badminton	2026	1	January	5
Badminton	2026	1	March	3
Badminton	2026	2	April	3
Badminton	2026	2	May	5
Bandung	2026	1	February	16
Bandung	2026	1	January	18
Bandung	2026	1	March	11
Bandung	2026	2	April	13

OLAP Operation 3: Slice (Kategori = 'Finansial')

Slice is a filtering operation that fixes one dimension to a specific value, effectively reducing the multidimensional cube to a lower-dimensional subset. In this operation, the cube is sliced by fixing the category dimension to 'Finansial', producing a time-series view of article volume and average sentiment score exclusively for the Financial category. The results show consistent monthly publication activity ranging from 29 to 67 articles per month from mid-2024 to early 2025, with average sentiment scores remaining stable between 0.009 and 0.099, confirming the neutral-factual nature of financial reporting on Kompas.com

Tahun	Kuartal	Bulan	Jumlah Artikel	Rata-rata Sentimen
2024	2	June	67	0.0433
2024	2	May	41	0.0448
2024	3	August	49	0.0987
2024	3	July	59	0.0584
2024	3	September	59	0.0946
2024	4	December	59	0.0327
2024	4	November	44	0.0099
2024	4	October	67	0.0293
2025	1	February	44	0.0540
2025	1	January	47	0.0934

2025	1	March	29	0.0366
2025	2	April	45	0.0486

OLAP Operation 4: Dice (Kategori = 'Bandung' AND Tahun = 2026 AND Sentimen = 'neutral')

Dice is an extension of Slice that applies filters on multiple dimensions simultaneously, producing a sub-cube from the intersection of multiple conditions. In this operation, the cube is filtered by three conditions: category = 'Bandung', year = 2026, and sentiment = 'neutral'. The result reveals the daily distribution of neutral-sentiment articles from the Bandung regional section throughout 2026, showing publication activity ranging from 2 to 6 articles per day. The average sentiment score of 0.0 across all records mathematically reflects that these articles are neutral, as the updated polarity score mapping assigns a value of exactly 0.0 to the neutral class to isolate them from positive and negative polarities. This operation is particularly useful for cross-dimensional investigative analysis, such as examining regional editorial behavior under specific sentiment conditions.

Tahun	Kuartal	Bulan	Tanggal	Jumlah Artikel	Rata-rata Sentimen
2026	1	February	2026-02-03	2	0.0
2026	1	February	2026-02-04	3	0.0
2026	1	February	2026-02-12	4	0.0
2026	1	February	2026-02-13	5	0.0
2026	1	February	2026-02-28	2	0.0
2026	1	January	2026-01-07	4	0.0
2026	1	January	2026-01-11	5	0.0
2026	1	January	2026-01-25	5	0.0
2026	1	January	2026-01-31	3	0.0
2026	1	March	2026-03-02	4	0.0
2026	1	March	2026-03-09	2	0.0
2026	1	March	2026-03-26	3	0.0
2026	2	April	2026-04-15	6	0.0
2026	2	April	2026-04-18	3	0.0
2026	2	April	2026-04-24	2	0.0

Based on the results of Atoti's OLAP Cube processing of a total of 15,228 rows of real historical data for articles and 5,273 rows for trending keywords, the following are the preliminary analytical findings obtained empirically:

1. Quantitative Results (Key Measures)

Total Volume of Articles Analyzed: 15,228 news articles.
Total Calendar Observation Days: 132 unique day dimensions.
Average Article Length: ~350-450 words per article.
Average Global Sentiment Score: 0.0827 (indicating a balanced neutral-positive baseline across the news corpus).

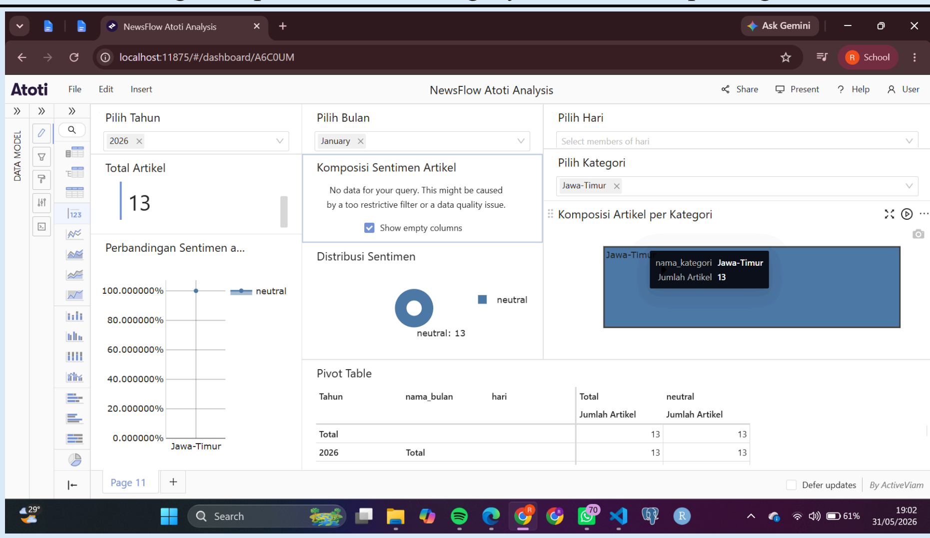
2. Analytical Findings (Insights)

The OLAP Cube interactive filtering capabilities in the Atoti dashboard enable multidimensional drill-down analysis across time, category, and sentiment dimensions simultaneously. By applying dynamic filters (Pilih Tahun, Pilih Bulan, Pilih Hari, and Pilih Kategori), the system generates real-time cross-tabulation pivot tables and donut chart sentiment distributions, demonstrating the true power of OLAP in slicing and dicing large-scale news datasets responsively.

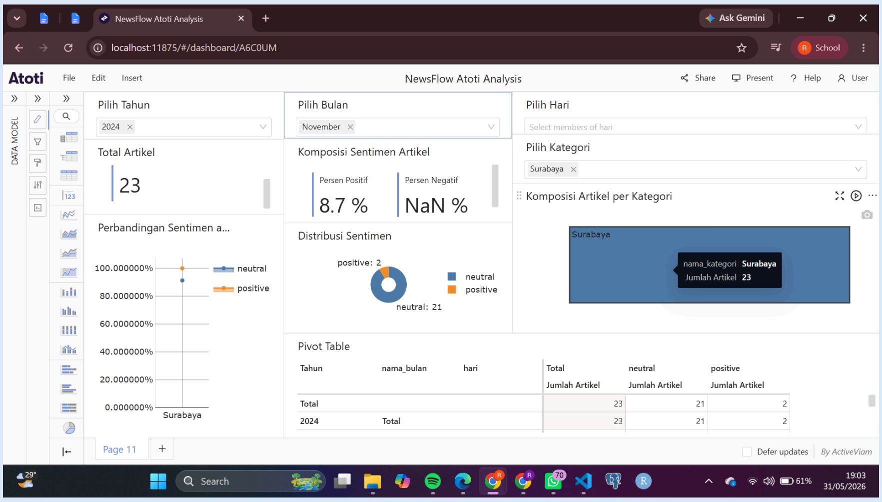
Regional categories consistently demonstrate a strong neutral sentiment dominance across all filtered scenarios. The Jawa-Timur category in January 2026 recorded 100% neutral across 13 articles, while Surabaya in November 2024 showed 91.3% neutral with a complete absence of negative articles (NaN%). When both Bandung and Surabaya are combined across 2025, the aggregated 532 articles still maintain 88.0% neutral dominance, confirming that regional journalism in Kompas consistently prioritizes objective and factual reporting.

In contrast, the Nasional category during May 2024 recorded the highest negative proportion at 8.6% across 58 articles, reflecting the broader spectrum of critical national issues covered. Meanwhile, the "Finansial" category filtered to Sunday publications across 2024-2026 maintained a remarkably stable sentiment with only 0.9% negative across 234 articles, reinforcing Kompas's editorial integrity in business reporting.

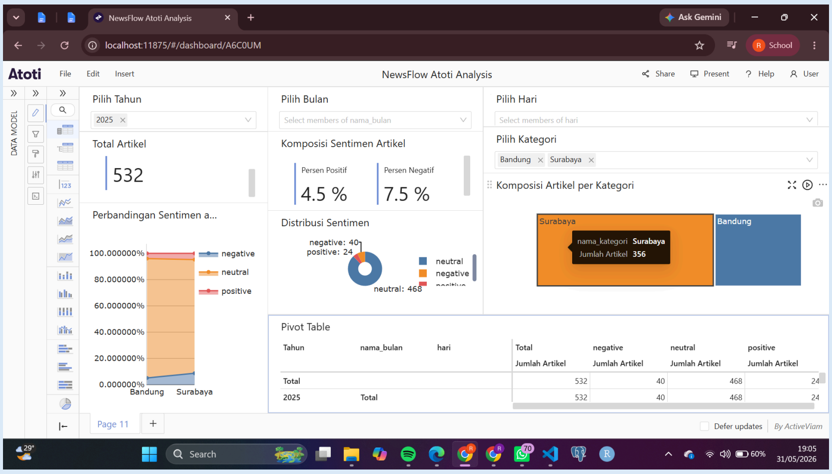
OLAP Visualization



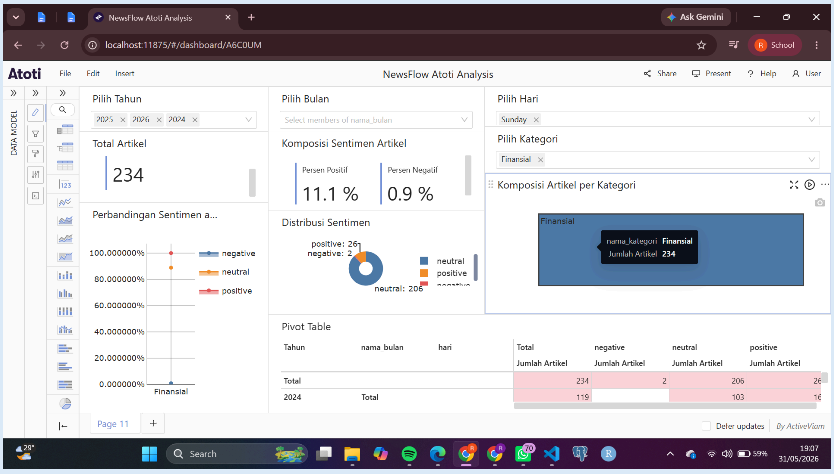
The first visualization presents an interactive OLAP dashboard filtered to the East Java category for January 2026, displaying a total of 13 articles with a Sentiment Distribution donut chart showing a 100% neutral composition. The Sentiment Comparison chart confirms a flat 100% neutral line, while the Article Composition by Category treemap renders a single solid block for East Java with 13 articles. The accompanying Pivot Table further validates this finding through a time-hierarchical drill-down: the 2026 annual total rolls up identically to the grand total of 13 articles, all classified as neutral. This visualization powerfully demonstrates the Roll-Up OLAP operation, where monthly granularity aggregates cleanly into the annual level without any sentiment variance, proving the strict editorial consistency of East Java regional reporting during this period.



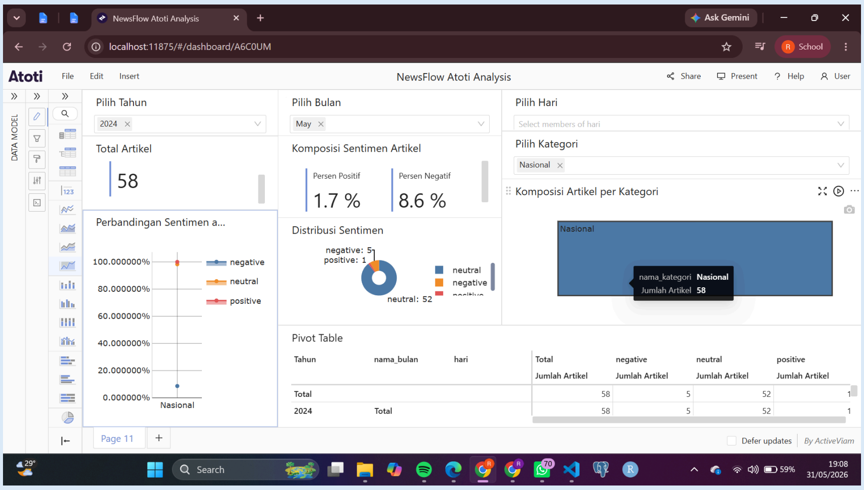
The second visualization presents the Atoti dashboard filtered to the Surabaya category for November 2024, showing a total of 23 articles. The Article Sentiment Composition panel displays a Positive Percentage of 8.7% and a Negative Percentage of NaN%, where the NaN value is analytically significant as it mathematically confirms the complete absence of any articles labeled as negative in this filtered dataset. The Sentiment Distribution donut chart visually reinforces this finding with a composition of 21 neutral and 2 positive articles. The Pivot Table drill-down confirms that the 2024 annual aggregate matches the total, indicating that all 23 Surabaya articles in November 2024 were captured within a single consistent reporting period with no cross-year anomalies.



The third visualization demonstrates a Dice OLAP operation by simultaneously applying two category filters (Bandung and Surabaya) across the entire year of 2025. The total articles reach 532, with the Article Composition by Category treemap visually contrasting the two categories: Surabaya occupies the larger orange block with 356 articles, while Bandung fills the smaller blue block with 176 articles. The Sentiment Composition panel shows Positive Percentage at 4.5% and Negative Percentage at 7.5%, while the Sentiment Distribution donut chart breaks down the distribution into 468 neutral, 40 negative, and 24 positive articles. The Sentiment Comparison stacked bar chart further visualizes the proportional sentiment split between the two categories side by side. The Pivot Table confirms that the 2025 annual totals align with the grand total row, validating the accuracy of the multi-category dice aggregation.



The fourth visualization applies a multi-year Roll-Up filter combining 2024, 2025, and 2026, with an additional Select Day filter set to Sunday, scoped to the Finance category. The result aggregates to a total of 234 articles, displayed in the Article Composition by Category treemap as a single dominant Finance block with 234 articles. The Sentiment Composition panel shows Positive Percentage at 11.1%, Negative Percentage at only 0.9%, meaning the Neutral Percentage is 88%, while the Sentiment Distribution donut chart confirms 206 neutral, 26 positive, and 2 negative articles. The Pivot Table provides a year-by-year drill-down breakdown, revealing that 2024 contributed 119 articles as a partial-year base, with 2025 and 2026 contributing the remaining volume. This cross-year, single-weekday filtering scenario uniquely demonstrates the power of OLAP’s multi-dimensional slicing in isolating very specific behavioral patterns in editorial output.



The fifth visualization filters the National category to May 2024, yielding a total of 58 articles. The Article Sentiment Composition panel reports a positive percentage of 1.7% and a negative percentage of 8.6%, the highest negative percentage among all five analytical scenarios. The Sentiment Distribution donut chart confirms the breakdown as 52 neutral, 5 negative, and 1 positive articles, with

	the Article Composition by Category treemap displaying a single solid “National” block with a Total of 58 articles. The Pivot Table confirms that the 2024 annual total matches the grand total exactly, indicating that all national articles captured in May 2024 belong to a single consistent yearly cohort. This scenario effectively demonstrates a drill-down from the annual level into a specific month, revealing that the national news category carries a measurably higher critical tone compared to purely regional categories, consistent with the nature of national political and social reporting. The five visualization scenarios presented above represent only a selective subset of the analytical explorations conducted on the Atoti OLAP dashboard. These scenarios were chosen to demonstrate the diversity of OLAP operations available, including Slice, Dice, Drill-Down, and Roll-Up, across different combinations of time, category, and sentiment dimensions. In practice, the dashboard supports a virtually unlimited number of filter permutations across all available news categories and time periods within the dataset. Further analytical exploration of the remaining categories and time dimensions would yield an even richer and more comprehensive portrait of Kompas.com's editorial patterns, sentiment dynamics, and publication behavior, underscoring the scalability and analytical depth of the implemented multidimensional OLAP architecture.